ARJUNA JEE 2027

Lecture -02 | Mathematics

SEQUENCE & SERIES

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Topics *to be covered*

A **Properties of A.P.**

B **Different types of Questions on A.P.**

C

D



Question

If second degree equation in x & y :

$$2x^2 - y^2 + xy - 3x + 9y + k = 0$$

Can be resolved as product of two linear factors, then find k, also find those linear factors.

Ans: $k = -20$

Factors: $x + y - 4 = 0$ & $2x - y + 5 = 0$

SEQUENCE & SERIES

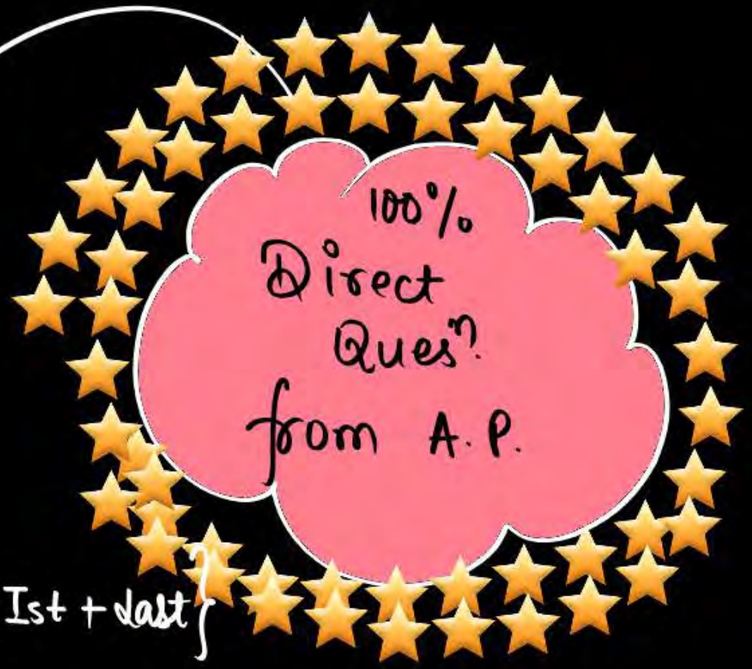
LAST CLASS REVISION

Answer the following Questions:

A.P. $\rightarrow a, a+d, a+2d, a+3d$ ... ...

1. n^{th} term of an A.P. is $T_n = a+(n-1)d$.
2. Sum of first n terms of an A.P. is _____.

$S_n = \frac{n}{2}\{2a+(n-1)d\} = \frac{n}{2}\{1st + last\}$



Question

For an A.P., $S_p = q$ and $S_q = p$, then S_{p+q} = _____.

A $p + q$

B $p - q$

C $q - p$

D $-p - q$

a, d

$S_p = \frac{p}{2}\{2a + (p-1)d\} = q \Rightarrow ap + (p-1)\frac{dp}{2} = q$

$S_q = \frac{q}{2}\{2a + (q-1)d\} = p \Rightarrow aq + (q-1)\frac{dq}{2} = p$

$\ominus$

$a(p-q) + \frac{d}{2}\{p^2 - p - q^2 + q\} = q - p$

$p^2 - q^2 - (p-q)$

$(p-q)(p+q) - (p-q)$

$(p-q)\{a + \frac{d}{2}(p+q-1)\} = -(p-q)$

$\frac{2a + d(p+q-1)}{2} = -1$

$S_{p+q} = \left(\frac{p+q}{2}\right)\{2a + (p+q-1)d\}$

$= \frac{(p+q)}{2}(-2) = -p - q$

Question (Tarika Casual ☆)

If sequence $< a_n >$ follows relation $a_n = 2a_{n-1} - a_{n-2}, n \geq 3$ and $a_1 = 7$ & $a_{10} = 30$, then $a_{40} =$ _____.

A.P.

$a_n = a_{n-1} + a_{n-1} - a_{n-2}$

$a_n - a_{n-1} = a_{n-1} - a_{n-2}$, $n \geq 3$

$a_1 \; a_2 \; a_3 \; a_4 \ldots \rightarrow$ A.P.

a, d

$a_1 = 7$ & $a_{10} = 30$

$a = 7$

$a + 9d = 30$

$d = \frac{23}{9}$

$n=3$ $a_3 - a_2 = a_2 - a_1$

$n=4$ $a_4 - a_3 = a_3 - a_2$

$n=5$ $a_5 - a_4 = a_4 - a_3$

$a_{40} = a + 39d = 7 + \frac{13}{\cancel{39}} \times \frac{23}{\cancel{9}_3}$

$= \frac{21 + 299}{3} = \frac{320}{3}$ Ans.

PROPERTIES OF A.P.

(i) For any sequence : $T_n = S_n - S_{n-1}$ ✓

(ii) For an A.P. : $d = S_n - 2S_{n-1} + S_{n-2}$ ✓

(iii) If $(a_1, a_2, a_3, \ldots\ldots a_n) \rightarrow$ A.P., then

$(a_1 \pm k, a_2 \pm k, a_3 \pm k, \ldots\ldots, a_n \pm k) \rightarrow$ A.P.

$(k a_1, k a_2, k a_3, \ldots\ldots, k a_n) \rightarrow$ A.P.

$a_1^2, a_2^2, a_3^2, \ldots. \rightarrow$ not in A.P.

$a_1a_2, a_2a_3, a_3a_4, \ldots\ldots \rightarrow$ not in A.P.

$\sqrt{a_1}, \sqrt{a_2}, \sqrt{a_3}\ldots. \rightarrow$ not in A.P.

9, 21, 33, 45 ..... → A.P. ✓

3 × ex:-

3, 7, 11, 15, 19, 23, 27 .... → A.P.

+2

5, 9, 13, 17, 21 ..... → A.P.

Sq.

9, 49, 121, 225 ..... → A.P. X

$S_n = \boxed{a_1 + a_2 + a_3 + \ldots\ldots + a_{n-1}} + a_n$

S_{n-1}

$S_n - S_{n-1} = a_n$

$S_{n-1} - S_{n-2} = a_{n-1}$

for A.P.

$d = a_2 - a_1 = a_3 - a_2 = a_4 - a_3 \ldots$

$d = a_n - a_{n-1}$

$= (S_n - S_{n-1}) - (S_{n-1} - S_{n-2})$



PROPERTIES OF A.P.

(iv) If a, b, c are in A.P. $\Leftrightarrow$ $2b = a + c$

$\Rightarrow$ $b - a = c - b$ OR

$2b = a + c$

2 (Middle) = Sum of other two.

RAPID FIRE ROUND

If $2^{1+x}, f(x)$ & 2^{1-x} are in A.P., then $f(3) =$ _____.

$$f(x) = \frac{2^{1+x} + 2^{1-x}}{2}$$

$x = 3$

$$f(3) = \frac{2^{1+3} + 2^{1-3}}{2} = \frac{16 + \frac{1}{4}}{2} = \frac{65}{8} \text{ Ans.}$$

PROPERTIES OF A.P.

(v) **3 terms in A.P.** $\rightarrow a - d, a, a + d$ → $a-d, d$.

4 terms in A.P. $\rightarrow a - 3d, a - d, a + d, a + 3d$ → $a-3d, 2d$

5 terms in A.P. $\rightarrow a - 2d, a - d, a, a + d, a + 2d$ → $a-2d, d$.

Above assumption is more useful when Sum of the terms is involved.

3 Terms in A.P. whose sum = 17.

$a, a+d, a+2d$

$3a + 3d = 17$ —(1)

$a - d + a + a + d = 17$

$3a = 17$

$a = \frac{17}{3}$.

Question [JEE Mains-2019]

If the sum and product of the first three term in an A.P. are 33 and 1155, respectively, then a value of its 11th term is

A. −25

B. 25

C. −36

D. −35

$a-d,\ a,\ a+d \rightarrow$ A.P.

$\text{Sum} = 3a = 33$

$a = 11$

$\text{Prod.} = (a-d)a(a+d) = 1155$

$a(a^2-d^2) = 1155$

$11(121-d^2) = 1155$ (105)

$121-d^2 = 105$

$16 = d^2$

$\pm 4 = d$

$a=11,\ d=4$

A.P. → 7, 11, 15

$T_{11} = a+10d$

$= 7+10\times 4$

$= 47$

$a=11,\ d=-4$

A.P. → 15, 11, 7

$T_{11} = a+10d$

$= 15+10(-4)$

$= 15-40 = -25$



PROPERTIES OF A.P.

(vi) k^{th} term from last = $(n - k + 1)^{th}$ term from starting

Total Terms

for every sequence

RAPID FIRE ROUND

(a) If $a_1, a_2, a_3, \ldots\ldots a_{77}$ are in A.P. then 35th term from last is 43rd term from starting.

(b) If first and last term of an A.P. with common difference 5 are 1 & 51 respectively. Then 4th term from last 36.

$(77 - 35 + 1)^{th}$ from st.

(b) $d = 5$

$a = 1$

$T_n = 51$

$1 + (n-1)5 = 51.$

$n - 1 = 10$

$n = 11$

4th from last = $(11 - 4 + 1)$ from st.

$= 8^{th}$

$T_8 = a + 7d$

$= 1 + 7 \times 5 = 36.$

RAPID FIRE ROUND

(a) If $a_1, a_2, a_3, \ldots\ldots a_{77}$ are in A.P. then 35th term from last is 43rd term from starting.

(b) If first and last term of an A.P. with common difference 5 are 1 & 51 respectively. Then 4th term from last 36.

$(77 - 35 + 1)^{th}$ from st.

A.P. → 1, 6, 11, 16, 21, 26, 31, 36, 41, 46, 51 ($d = 5$)

(M-2) 51, 46, 41, 36, ... → A.P. ($d = -5$)

51 → Ist

$T_4 = a + (4-1)d$

$= 51 + 3 \times (-5)$

$= 51 - 15 = 36$

PROPERTIES OF A.P.

(vii) Sum of terms of an A.P. equidistant from starting and end is equal .

$a_1, a_2, a_3, a_4, a_5 \ldots\ldots a_{37} \rightarrow$ A.P.

$$a_1 + a_{37} = a_2 + a_{36} = a_3 + a_{35} = \ldots\ldots$$

$a + a + 36d$ $\quad$ $a + d + a + 35d$ $\quad$ $a + 2d + a + 34d$

$2a + 36d$.

Simple.

$a_1, a_2, a_3 \ldots\ldots a_{149}$ → A.P.

$a_{121} + a_3 = a_{100} + a_{24}$

$= a_{102} + a_{22}$.

$a_5 + a_{11} = a_6 + a_{10} = a_2 + a_{14} = a_8 + a_8 = a_4 + a_{12}$

$a+4d + a+10d$ $\quad$ $a+5d + a+9d$

$a_{11} + a_{27} = a_8 + a_{30} = a_{20} + a_{18} = a_{19} + a_{19}$

Question [JEE Mains-2019]

If $a_1, a_2, a_3, \ldots, a_n$ are in A.P. and $a_1 + a_4 + a_7 + \cdots + a_{16} = 114$, then $a_1 + a_6 + a_{11} + a_{16}$ is equal to

A. 98

B. 38

C. 64

D. 76

$a_1 + a_4 + a_7 + a_{10} + a_{13} + a_{16} = 114$

$a + a+3d + a+6d + a+9d + a+12d + a+15d = 114$

$6a + 45d = 114$

$(2a + 15d) = \frac{114}{3}$

$a + a+5d + a+10d + a+15d$

$4a + 30d$

$2(2a + 15d)$

$2 \times \frac{114}{3} = 76$

Question [JEE Mains-2019]

$a_1 + a_{16} = K$

If $a_1, a_2, a_3, \ldots, a_n$ are in A.P. and $a_1 + a_4 + a_7 + \cdots + a_{16} = 114$, then $a_1 + a_6 + a_{11} + a_{16}$ is equal to

A 98

B 38

C 64

D 76

$a_1 + a_4 + a_7 + a_{10} + a_{13} + a_{16} = 114$.

$K + K + K = 114$

$3K = 114$

$K = 38$.

$a_1 + a_6 + a_{11} + a_{16}$

$K + K = 2K$

$= 76$.



PROPERTIES OF A.P.

(viii) If $d > 0 \Rightarrow$ Increasing A.P.
If $d < 0 \Rightarrow$ Decreasing A.P.
If $d = 0 \Rightarrow$ Constant A.P.

(ix) **For an A.P. :**

General Term = T_n = Linear function of 'n'
Sum of 'n' terms = S_n = Quadratic function of 'n' with (constant term = 0)

(x) If $a_1, a_2, a_3, \ldots \rightarrow$ A.P.

and $b_1, b_2, b_3, \ldots \rightarrow$ another A.P.

Then : $a_1 \pm b_1, a_2 \pm b_2, a_3 \pm b_3, \ldots \rightarrow$ A.P.

RAPID FIRE ROUND

If n^{th} term of are A.P. is $T_n = 7n + 11,\ n \geq 1$ then:

(i) first term (a) = 18.

(ii) common difference (d) = 7.

$n = 1$

$T_1 = 7 + 11$

$= 18$

$d = T_2 - T_1$

$= 7$

$n = 2 \rightarrow T_2 = 14 + 11$

$= 25$

$T_n = a + (n-1)d$

$= a + nd - d$

$T_n = (d)n^1 + (a-d)$

$d = 7$

$a - d = 11$

$a = 18$

RAPID FIRE ROUND

If sum of first n terms of an A.P. is $S_n = 3n^2 + 5n$ then

(i) first term (a) = $S_1 = 8$.

(ii) common difference (d) = 6.

(iii) n^{th} term (T_n) = $6n+2$.

$S_1 = T_1 = a$

$n=2 \Rightarrow S_2 = 3\times 2^2 + 5\times 2$.

$= 12 + 10$.

$T_1 + T_2 = 22$

$8 + T_2 = 22$.

$T_2 = 14$.

$d = 14 - 8$

$d = 6$

$T_n = a + (n-1)d$.

$= 8 + (n-1)6$.

$= 6n + 2$.

RAPID FIRE ROUND

If sum of first n terms of an A.P. is $S_n = 3n^2 + 5n$ then

(i) first term (a) = $S_1 = 8$. ✓.

(ii) common difference (d) = 6.

(iii) n^{th} term (T_n) = $6n+2$.

$S_1 = T_1 = a$

$$S_n = \frac{n}{2}\{2a + nd - d\}$$

$$= na + \frac{n^2 d}{2} - \frac{nd}{2}$$

$$\# \quad S_n = \left(\frac{d}{2}\right)n^2 + \left(a - \frac{d}{2}\right)n^1$$

$\frac{d}{2} = 3$ $\quad a - \frac{d}{2} = 5$.

$d = 6$. $\quad a = 8$.

$$T_n = S_n - S_{n-1}$$

$$= 3n^2 + 5n - (3(n-1)^2 + 5(n-1))$$

$$= 3n^2 + 5n - (3n^2 + 3 - 6n + 5n - 5)$$

$$= 3n^2 + 5n - 3n^2 + 2 + n$$

$$= 6n + 2.$$

QUESTION PRACTICE

Question [JEE Mains-2023]

$a = 8$, $d = 3$, $n = 14$

Let $a_1 = 8$ $a_2, a_3, \ldots a_n$ be an A.P. If the sum of its first four terms is 50 and the sum of its last four terms is 170, then the product of its middle two terms is

$a_1 + a_2 + a_3 + a_4 = 50$

$8 + 8+d + 8+2d + 8+3d = 50$

$32 + 6d = 50$

$6d = 18$

$d = 3$

$a_n + a_{n-1} + a_{n-2} + a_{n-3} = 170$

$a+(n-1)d + a+(n-2)d + a+(n-3)d + a+(n-4)d = 170$

$4 \times 8 + 3\{4n-10\} = 170$

$12n = 168$

$n = 14$

7th & 8th

$(8+6\times3) \times (8+7\times3) = 754$

Middle Term :-

$$\frac{n}{2}, \quad \frac{n+1}{2}, \quad \frac{n+2}{2}$$

$$\boxed{\frac{\text{Total Terms}}{2}, \quad \frac{\text{Total Terms}+1}{2}, \quad \frac{\text{Total Terms}+2}{2}}$$

Total Terms = 14 $\longrightarrow$ $\frac{14}{2}^{th}$, $\frac{14}{2}$, $\left(\frac{14}{2}+1\right)^{th}$

Total Terms = 51 $\longrightarrow$ $\frac{51}{2}$, $\frac{51+1}{2}$, $\frac{51+2}{2}$

$\frac{51+1}{2} = 26^{th}$.

Total Terms = 144

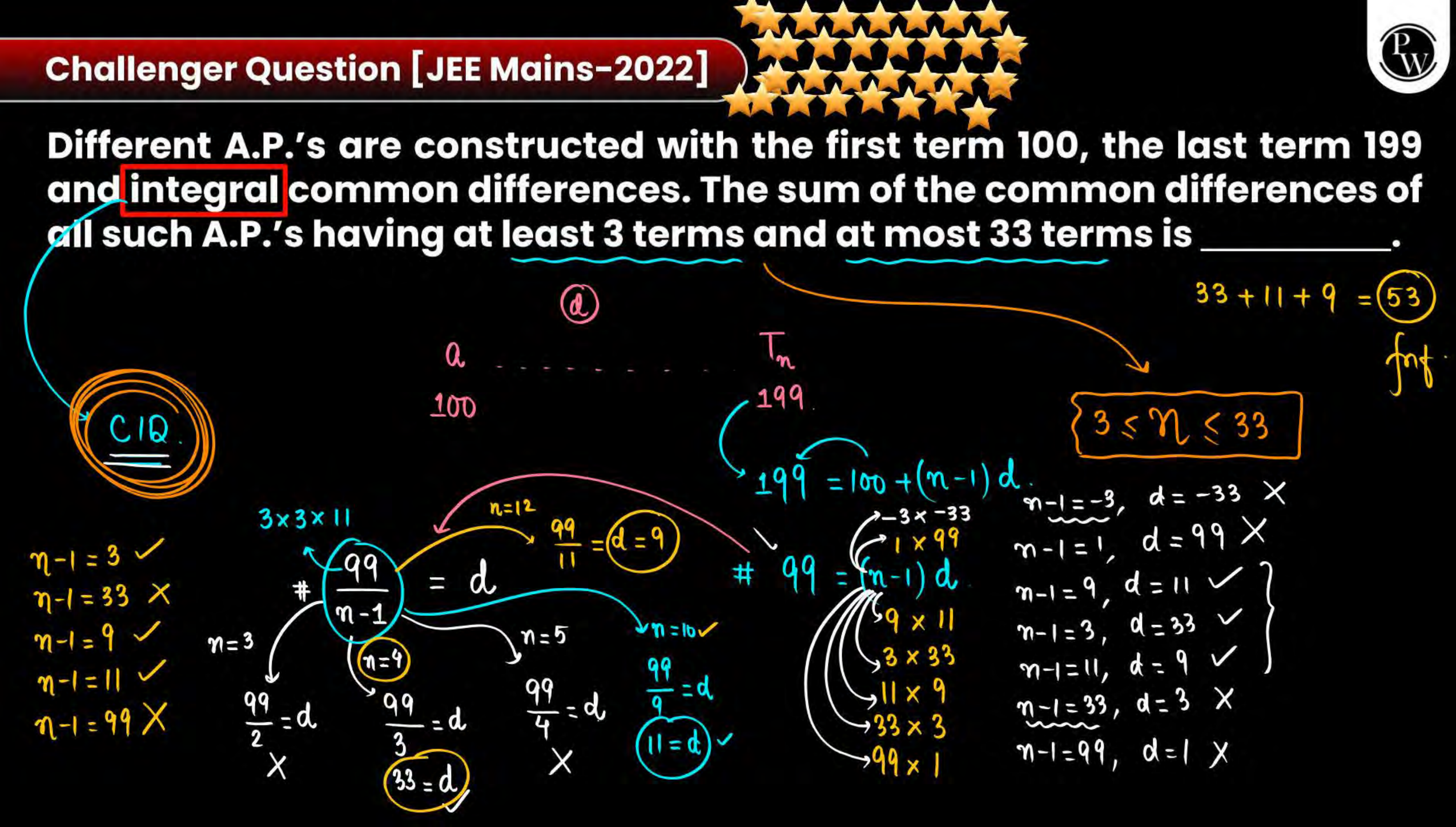
$\frac{144}{2}$, $\frac{144+1}{2}$, $\frac{144+2}{2}$

$\frac{144}{2} = 72^{nd}$.

$\frac{144+2}{2} = 73^{rd}$.

Challenger Question [JEE Mains-2022]

Different A.P.'s are constructed with the first term 100, the last term 199 and integral common differences. The sum of the common differences of all such A.P.'s having at least 3 terms and at most 33 terms is _____.



Question [JEE (Adv.)-2015]

Suppose that all the terms of an arithmetic progression (A.P.) are natural numbers. If the ratio of the sum of the first seven terms to the sum of the first eleven terms is 6 : 11 and the seventh term lies in between 130 and 140, then the common difference of this A.P. is

$a_1, a_2, a_3 \ldots \rightarrow$ A.P. (N.)

$$\frac{S_7}{S_{11}} = \frac{6}{11}$$

$$\frac{\frac{7}{2}\{2a+6d\}}{\frac{11}{2}\{2a+10d\}} = \frac{6}{11}$$

$14a + 42d = 12a + 60d$

$2a = 18d$

$a = 9d$

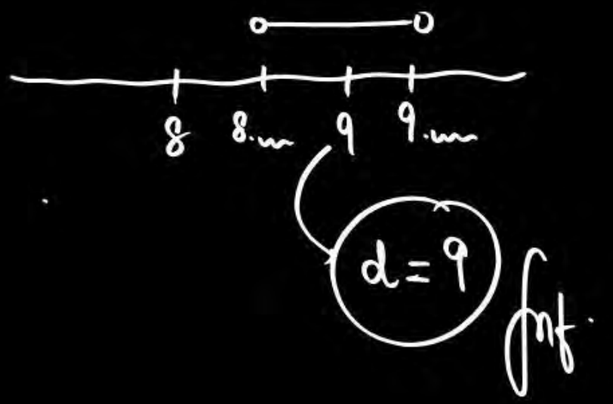
$130 < a_7 < 140$

$130 < a + 6d < 140$

$130 < 15d < 140$

$\frac{130}{15} < d < \frac{140}{15}$

$8.\overline{m} < d < 9.\overline{m}$



Question [JEE Mains-2022]

Suppose $a_1, a_2, \ldots, a_n, \ldots$ be an arithmetic progression of natural numbers. If the ratio of the sum of first five terms to the sum of first nine terms of the progression is $5 : 17$ and $110 < a_{15} < 120$, then the sum of the first ten terms of the progression is equal to

A 290

B 380

C 460

D 510

Q iBY – (A)

Chat!!

Question [JEE Mains-2025]

Suppose that the number of terms in an A.P. is $2k, k \in N$. If the sum of all odd numbered terms of the A.P. is 40, the sum of all even numbered terms is 55 and the last term of the A.P. exceeds the first term by 27, then k is equal to:

A. 6

B. 4

C. 5

D. 8

$a_1, a_2, a_3, a_4 \ldots a_{2k-1}, a_{2k}$ → A.P. (2k terms)

$a_1 + a_3 + a_5 + a_7 + \ldots + a_{2k-1} = 40$ — ①

$a_2 + a_4 + a_6 + \ldots + a_{2k} = 55$ — ②

① − ② (k terms): $-d + (-d) + (-d) + \ldots = -15$

$a_{2k} = a_1 + 27$

$a + (2k-1)d = a + 27$

$2kd - d = 27$

$30 - d = 27$

$3 = d$

$-dk = -15$

$dk = 15$

$k = 5$

Question [JEE Mains-2025]

The number of terms of an A.P. is even; the sum of all the odd terms is 24, the sum of all the even terms is 30 and the last term exceeds the first by $\frac{21}{2}$. Then the number of terms which are integers in the A.P. is:

A 4

B 10

C 6

D 8

Q1 BY – (B)

Chat.

DO IT BY YOURSELF (DIBY)

DIBY-01

Let a_n be the nth term of an $A.P.$ If $S_n = a_1 + a_2 + a_3 + \cdots . + a_n = 700, a_6 = 7$ and $S_7 = 7$, then a_n is equal to: [JEE Mains-2025]

(A) 56 (B) 65 (C) 64 (D) 70

[Ans. : (C)]

DIBY-02

If the first term of an A.P. is 3 and the sum of its first four terms is equal to one-fifth of the sum of the next four terms, then the sum of the first 20 terms is equal to [JEE Mains-2025]

(A) –120 (B) –1200 (C) –1020 (D) –1080

[Ans. : (D)]

DIBY-03

The roots of the quadratic $3x^2 - px + q = 0$ are 10th and 11th terms of an arithmetic progression with common difference $\frac{3}{2}$. If the sum of the first 11 terms of this arithmetic progression is 88, then $q - 2p$ is equal to [JEE Mains-2025]

[Ans. : 474]

DIBY-04

In an arithmetic progression, if $S_{40} = 1030$ and $S_{12} = 57$, then $S_{30} - S_{10}$ is equal to

[JEE Mains-2025]

(A) 505 (B) 525 (C) 510 (D) 515

[Ans. : (D)]

DIBY-05

If $a_1, a_2, a_3, \ldots$ are in A.P. such that $a_1 + a_7 + a_{16} = 40$, then the sum of the first 15 terms of this A.P. is

[JEE Mains-2019]

(A) 200 (B) 280 (C) 120 (D) 260

[Ans. : (A)]

DIBY-06

If 19th terms of non-zero A.P. is zero, then its $(49^{th}$ term$) : (29^{th}$ term$)$ is:

[JEE Mains-2019]

(A) 4 : 1 (B) 1 : 3 (C) 3 : 1 (D) 2 : 1

[Ans. : (C)]

DIBY-07

Let S_n be the sum of the first n terms of an arithmetic progression. If $S_{3n} = 3S_{2n}$, then the value of $\frac{S_{4n}}{S_{2n}}$ is:

[JEE Mains-2021]

(A) 2 (B) 6 (C) 8 (D) 4

[Ans. : (B)]

DIBY-08

Let $a_1, a_2, a_3, \ldots$ be an A.P. If $\frac{a_1+a_2+\cdots+a_{10}}{a_1+a_2+\cdots+a_p} = \frac{100}{p^2}, p \neq 10$, then $\frac{a_{11}}{a_{10}}$ is equal to:

[JEE Mains-2021]

(A) $\frac{21}{19}$ (B) $\frac{100}{121}$ (C) $\frac{121}{100}$ (D) $\frac{19}{21}$

[Ans. : (A)]

DIBY-09

If $\log_3 2, \log_3(2^x - 5), \log_3\left(2^x - \frac{7}{2}\right)$ are in an arithmetic progression, then the value of x is equal to

[JEE Mains-2021]

[Ans. : 3]

DIBY-10

If the 10th term of an A.P. is $\frac{1}{20}$ and its 20th term is $\frac{1}{10}$, then the sum of its first 200 terms is

[JEE Mains-2020]

(A) 100 (B) $50\frac{1}{4}$ (C) $100\frac{1}{2}$ (D) 50

[Ans. : (C)]



Homework

Re-attempt all Class Examples

DPP & DIBY

Module:

Exercise-02(Prabal): COMPLETE ✓

Exercise-03(Setu): COMPLETE ✓

It is never about end result

It's always about Journey

Thank You

#futureIITians